
GNN-Based BERT for Understanding Context from Music project

A Hybrid Graph Neural Network and BERT Approach to Multi-Label Music Tag Understanding for CSE425

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Abstract

1 Music context understanding requires modeling both the structural properties of au-
2 dio (harmonic and timbral patterns across time) and the semantic content expressed
3 through language (tags, captions, and descriptions). This project implements a
4 three-stage pipeline that progressively builds toward this goal: (1) a BERT-based
5 multi-label tag classifier trained on natural-language captions, (2) a Graph Neural
6 Network (GraphSAGE and GAT variants) that predicts genre-related tags directly
7 from audio structure graphs built from mel-spectrogram segments, and (3) a cross-
8 attention fusion model that combines both representations. For this project we
9 have used the MagnaTagATune dataset for audio and tag supervision, and Music-
10 Caps for caption-based supervision, and shown that the fusion model substantially
11 outperforms either single-modality baseline, achieving a Macro-F1 of 0.64 versus
12 0.37 for the GNN-only baseline and 0.35 for a GAT-only variant, while the stan-
13 dalone BERT caption classifier reaches 0.67 Macro-F1 on its own caption-to-tag
14 task. These results support the central hypothesis that audio structure and textual
15 semantics combined together cover more ground for far more accurate predictions.

16 1 Project Motivation

17 Music is a multi-layered signal where context spans melody, harmony, rhythm, timbre, and subjective
18 human semantics such as mood or genre. Purely sequential audio models (CNNs or RNNs over
19 spectrograms) capture local acoustic patterns but do not explicitly represent the relational structure
20 of a track, that is, how one segment relates to another, or how a harmonic pattern co-occurs with a
21 semantic tag. This project addresses that gap by combining a Graph Neural Network, which models
22 audio as a graph of temporally and harmonically related segments, with a BERT text encoder, which
23 captures semantic descriptions of the same audio. The two modalities are fused via cross-attention,

24 allowing the model to train its structural understanding of a track on its textual description, and vice
25 versa.

26 **2 Datasets**

27 Two datasets were used, MagnaTagATune and MusicCaps were chosen for this project.

28 **2.1 MagnaTagATune**

29 MagnaTagATune provides 25,863 audio clips (~30s each) paired with a 188-dimensional binary tag
30 annotation matrix covering genre, instrumentation, mood, and vocal characteristics. From the full
31 188-tag vocabulary, a genre-relevant subset of 10 tags was selected as the prediction target for the
32 GNN and fusion models for this project: classical, rock, jazz, electronic, pop, ambient, metal, folk,
33 country, and techno. The remaining ~178 tags (timbral, instrumental, and descriptive tags such as
34 “guitar”, “female voice”, “fast”, “mellow”) were used to construct a pseudo-caption text string per clip
35 for the BERT side of the fusion model, since MagnaTagATune does not provide free-text captions
36 directly.

37 **2.2 MusicCaps**

38 MusicCaps provides 5,521 clips with curated captions and an aspect list of short descriptive phrases,
39 sourced from Google’s public release. This dataset was used exclusively for the BERT model in task
40 1, framed as a caption-to-tag proxy task: the caption text is the model input, and the top-50 most
41 frequent aspect list phrases form the multi-label prediction target.

42 **2.3 Dataset Design Rationale**

43 An initial plan combined MagnaTagATune with the FMA dataset for cleaner single-label genre
44 supervision. This was reconsidered: FMA lacks paired text/caption data for the same clips, which
45 would have prevented a coherent Task 3 fusion story. MagnaTagATune alone satisfies the project’s
46 dataset requirement as it independently provides both an audio and a tag/text modality, and ensures
47 that Tasks 2 and 3 operate on directly comparable, paired data, which provided a stronger basis for
48 the ablation comparisons in Section 5.

49 **3 Preprocessing Pipeline**

50 **3.1 Audio Feature Extraction**

51 All audio was resampled to 22,050 Hz. Two feature representations were extracted per track using
52 librosa:

- 53 • **Log-mel spectrogram** (128 mel bins), capturing timbral and instrumentation-related infor-
54 mation.
- 55 • **Chroma features** (12 bins), capturing harmonic/pitch-class content.

56 Each track was segmented into fixed 5-second windows, discarding any trailing incomplete segment.
57 A 30-second MagnaTagATune clip typically yields 5 segments.

58 **3.2 Graph Construction**

59 For each track, a segment graph was constructed with one node per time segment. Two edge types
60 were added:

- 61 • **Temporal edges** connecting consecutive segments (weight = 1.0).
- 62 • **Similarity edges** connecting non-adjacent segments whose pooled feature vectors have
63 cosine similarity above $\tau = 0.8$.

Node features are the mean-pooled feature vector (mel or chroma) for that segment. The resulting graph is converted into a PyTorch Geometric Data object (node feature matrix, edge index, edge weights) for GNN consumption.

3.3 Text Preprocessing

Captions (MusicCaps) and pseudo-captions (MagnaTagATune non-genre tags, concatenated into a space-separated string) were tokenized using the bert-base-uncased tokenizer with a maximum sequence length of 64–128 tokens, with padding and truncation applied uniformly.

3.4 Train/Validation Splits

A single 80/20 clip-level split was generated once (based on clip_id) and reused identically across Task 2 and Task 3, ensuring both models are evaluated on the same held-out clips and that reported improvements from fusion are not an artifact of differing evaluation sets.

4 Model Architectures and Tasks

4.1 Task 1: BERT Baseline for Caption-to-Tag Understanding

A bert-base-uncased encoder produces a CLS embedding for each caption, passed through a linear classification head with sigmoid activation, trained with binary cross-entropy over the top-50 MusicCaps aspect tags:

$$\begin{aligned} t &= \text{BERT}_{\text{CLS}}(X_{\text{text}}) \\ \hat{y} &= \sigma(W^T t + b) \\ \mathcal{L}_{\text{BERT}} &= -\frac{1}{K} \sum_k [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] \end{aligned}$$

Trained for 5 epochs, batch size 8, learning rate 2e-5, on an 80/20 split of MusicCaps (~4,400 / ~1,100 clips).

4.2 Task 2: GNN on Music Structure Graphs

Two graph encoder variants were implemented and compared, both consuming mel-spectrogram-based segment graphs and predicting the 10-tag genre subset via mean-pooled graph-level classification:

- **GraphSAGE**: two SAGEConv layers (hidden dim 128) with mean-pool readout.
- **GAT**: two GATConv layers (hidden dim 128, 4 attention heads) with mean-pool readout.

$$\begin{aligned} h_i^{(l+1)} &= \sigma\left(W^{(l)} \cdot \text{AGG}(h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\})\right) \\ g &= \text{MEAN}_i(h_i^L) \\ \hat{y} &= \sigma(Wg + b) \end{aligned}$$

Both trained for 15 epochs, batch size 16, Adam optimizer (lr = 1e-3), with the best validation macro F1 checkpoint retained rather than the final epoch.

4.3 Task 3: GNN-BERT Fusion via Cross-Attention

The fusion model encodes the same mel-based segment graph with a GraphSAGE encoder (graph embedding g) and the pseudo-caption text with BERT (token embeddings H_{text}). A single-query cross-attention layer lets the graph embedding attend over the text token embeddings:

$$Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad V = H_{\text{text}}W_V$$

98

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)$$

99

$$\text{attended} = AV$$

100

$$z = \text{CONCAT}(g, \text{attended})$$

101

$$\hat{y} = \sigma(Wz)$$

102 Trained for 10 epochs, batch size 8, AdamW optimizer (lr = 2e-5, since BERT parameters are
103 fine-tuned jointly), with best-checkpoint selection identical to Task 2.

104 5 Results

105 5.1 Task 1: BERT Caption-to-Tag Classification

Table 1: Task 1 (BERT, MusicCaps caption $\rightarrow$ top-50 tag) training curve. Best Macro-F1: 0.667.

Epoch	Train Loss	Val Loss	Macro-F1	Micro-F1
1	0.2095	0.1412	0.264	0.482
2	0.1253	0.1100	0.422	0.624
3	0.1006	0.0962	0.518	0.684
4	0.0851	0.0871	0.622	0.727
5	0.0732	0.0818	0.667	0.744

106 5.2 Task 2: GNN Baselines (GraphSAGE vs. GAT, chroma vs. mel)

107 An initial run using chroma features (12-dim, harmonic content only) achieved a peak Macro-F1 of
108 only 0.16, motivating a switch to mel-spectrogram features (128-dim, timbral content), which nearly
109 doubled performance.

Table 2: Task 2 ablation across feature type and graph encoder. (*chroma result from an earlier 600-sample pilot run, included for the feature-type ablation only.)

Model	Node Features	Macro-F1	Micro-F1
GraphSAGE	Chroma (12-dim)	0.16*	0.35*
GraphSAGE	Mel (128-dim)	0.37	0.59
GAT	Mel (128-dim)	0.35	0.59

110 5.3 Task 3: GNN-BERT Fusion

Table 3: Task 3 comparison on the shared MagnaTagATune genre-tag prediction task, same 80/20 split as Table 2.

Model	Macro-F1	Micro-F1
GraphSAGE (GNN-only)	0.37	0.59
GAT (GNN-only)	0.35	0.59
Fusion (GraphSAGE + BERT, cross-attention)	0.64	0.73

111 Fusion improves Macro-F1 by +0.27 (a 73% relative improvement) and Micro-F1 by +0.14 over the
112 best GNN-only baseline. This is consistent with the project’s central hypothesis: harmonic/timbral
113 structure and tag-derived semantic description carry complementary signal for genre-style clas-
114 sification, and jointly attending over both yields a materially stronger representation than either
115 alone.

116 6 Discussion

117 6.1 Feature choice matters more than graph architecture

118 The single largest improvement within Task 2 came from switching node features from chroma to mel-
119 spectrogram (Macro-F1 0.16 to 0.37), not from the choice between GraphSAGE and GAT (0.37 vs.
120 0.35, within noise of each other). This suggests that for genre-style prediction, timbral/instrumentation
121 information carried in mel-spectrograms is substantially more discriminative than the purely harmonic
122 information in chroma, and that the specific message-passing scheme is a secondary factor once
123 informative node features are available.

124 6.2 Fusion provides the largest overall gain

125 The jump from GNN-only (0.37) to Fusion (0.64) is far larger than the gap between GNN variants,
126 indicating that textual/tag-derived semantic information contributes information genuinely orthogonal
127 to what audio structure alone provides. This is the expected outcome under the assignment’s core
128 premise, and the magnitude of the gap here is a stronger piece of evidence for that premise than either
129 single-modality result in isolation.

130 6.3 Limitations

- 131 • The pseudo-caption text used in Task 3 (concatenated MagnaTagATune tags) is not natural
132 language; it is a proxy chosen because MagnaTagATune does not provide free-text captions.
133 A natural next step is using true captions matched to audio for a more realistic fusion setting.
- 134 • Genre prediction is treated as a 10-tag multi-label proxy since FMA lacks paired text data.
135 This was a deliberate trade-off in favor of a coherent fusion model over label cleanliness.
- 136 • Training set sizes (1,600–2,000 samples) are a fraction of the full MagnaTagATune corpus
137 (25,863 clips); results are expected to improve further with the full dataset given additional
138 compute time.

139 7 Conclusion

140 This project implemented and evaluated three of the four tasks specified in the assignment: a BERT-
141 based text-to-tag classifier, a GNN-based audio-structure-to-tag classifier (with a GraphSAGE vs.
142 GAT ablation), and a cross-attention GNN–BERT fusion model. Across all experiments, the fusion
143 approach substantially outperformed any single-modality baseline (Macro-F1 0.64 vs. 0.37 for the
144 best GNN-only model), directly supporting the assignment’s motivating claim that music context
145 understanding benefits from jointly modeling structural and semantic information.

146 8 Reproducibility

147 All code, the exact train/validation split (data/splits/), and instructions to reconstruct the raw data
148 directories from public sources are available in the project’s GitHub repository. Model checkpoints
149 (.pt files) are excluded from version control due to size but can be regenerated by re-running the
150 corresponding training script and each script prints the metrics reported in this report.